

Weihaio Lu

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🌐 <https://luweihaio.github.io/>



Employment History

- 2026.2 – **Research Fellow**
Department of Statistics and Data Science, National National University of Singapore.
Principal Investigator: Dongming Huang.
- 2025.2 – 2026.2 **Research Fellow**
Department of Statistics and Data Science, National National University of Singapore.
Principal Investigator: Yingcun Xia.

Education

- 2020.9 – 2024.12 **Ph.D. in Statistics**
Department of Statistics and Data Science, Tsinghua University.
Advisor: Qian Lin.
- 2015.9 – 2019.6 **B.S. in Statistics**
School of Mathematics and Statistics, Wuhan University.

Research Interest

Large-dimensional spectral algorithms, minimax risks in non-parametric regression.

Research Publications

Journal Articles

- 1 Haobo Zhang, Yicheng Li, **Weihaio Lu**, and Qian Lin, “Optimal rates of kernel ridge regression under source condition in large dimensions,” *Journal of Machine Learning Research*, 2025.
- 2 Haobo Zhang, **Weihaio Lu**[†], and Qian Lin, “The phase diagram of kernel interpolation in large dimensions,” *Biometrika*, 2024.

Conference Proceedings

- 1 **Weihaio Lu**, Haobo Zhang, Yicheng Li, and Qian Lin, “On the saturation effects of spectral algorithms in large dimensions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- 2 Haobo Zhang, Yicheng Li, **Weihaio Lu**, and Qian Lin, “On the optimality of misspecified kernel ridge regression,” in *International Conference on Machine Learning (ICML)*, 2023.

Preprints

- 1 **Weihaio Lu**, Qian Lin, Yingcun Xia, and Dongming Huang, *Learning curves and benign overfitting of spectral algorithms in large dimensions*, 2026.

- 2 **Weihao Lu**, Jialin Ding, Haobo Zhang, and Qian Lin, *On the pinsker bound of inner product kernel regression in large dimensions*, 2024.
- 3 **Weihao Lu**, Haobo Zhang, Yicheng Li, Manyun Xu, and Qian Lin, *Optimal rate of kernel regression in large dimensions*, 2023.

Awards

- 2023
 - 📌 **Top Ten Papers Award**, Ninth National Graduate Statistics Forum, Renmin University of China.
 - 📌 **Best Poster Award**, Peking-Tsinghua Joint Statistics Colloquium, Peking University.
 - 📌 **First Prize**, Seventh National Statistics Doctoral Academic Forum, Guangzhou University.
 - 📌 **Second Prize for Academic Papers**, First National Doctoral Academic Forum on Mathematics and Statistics, Liaoning University.
- 2024
 - 📌 **NeurIPS 2024 Scholar Award**
- 2025
 - 📌 **AISTATS 2025 Best Reviewer Award**
 - 📌 **NeurIPS 2025 "Reviewer : Top Reviewer" Award**

Teaching Assistants

- 2020 Fall and 2021 Fall
 - 📌 Elementary Probability Theory.
- 2021 Spring
 - 📌 Reliability Data and Survival Analysis.
- 2022 Spring, 2023 Spring, and 2024 Spring
 - 📌 Mathematical Statistics.

Academic Services

- 📌 Conference Reviewer NeurIPS, 2025
- 📌 Conference Reviewer AISTATS, 2025, 2026